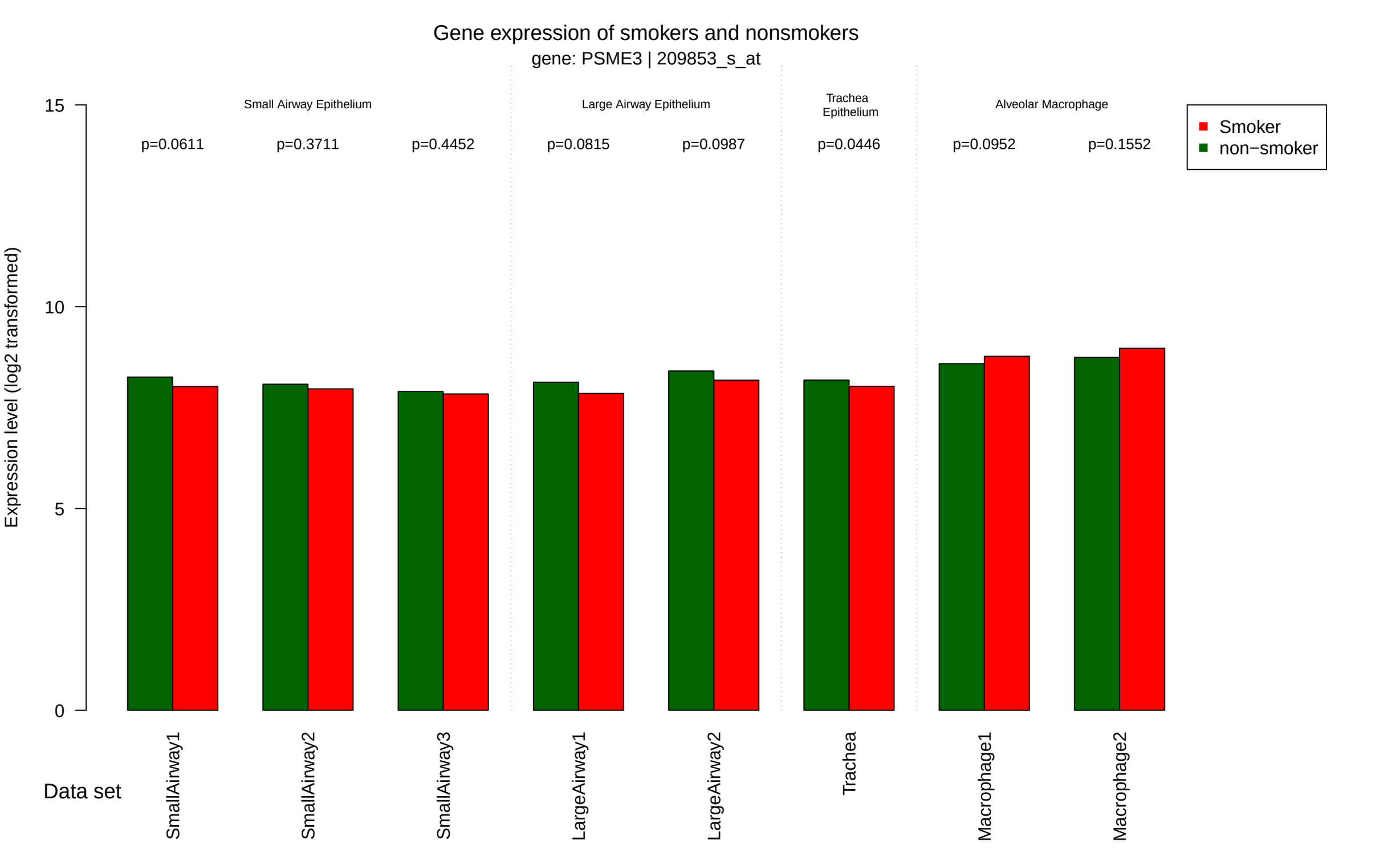
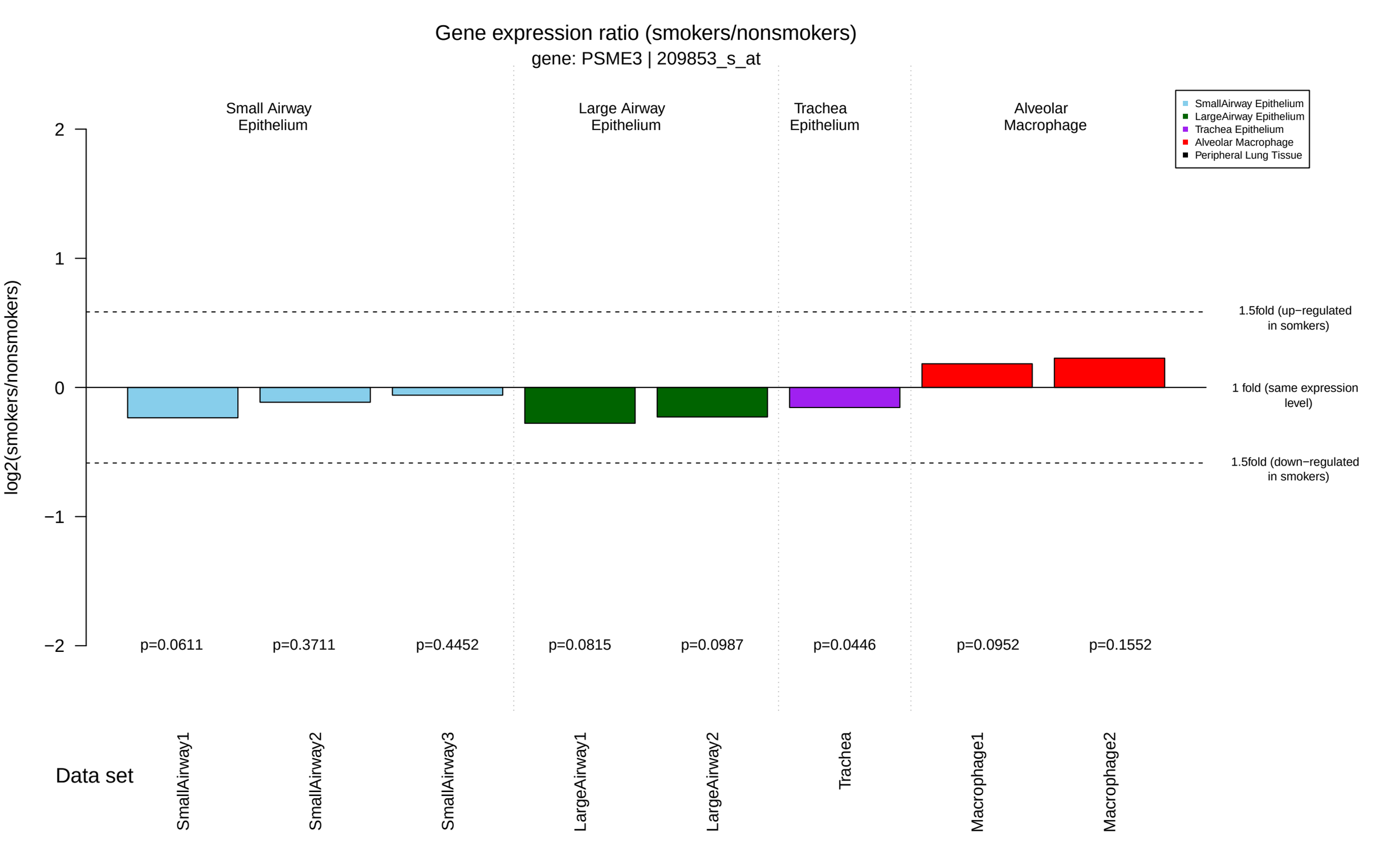
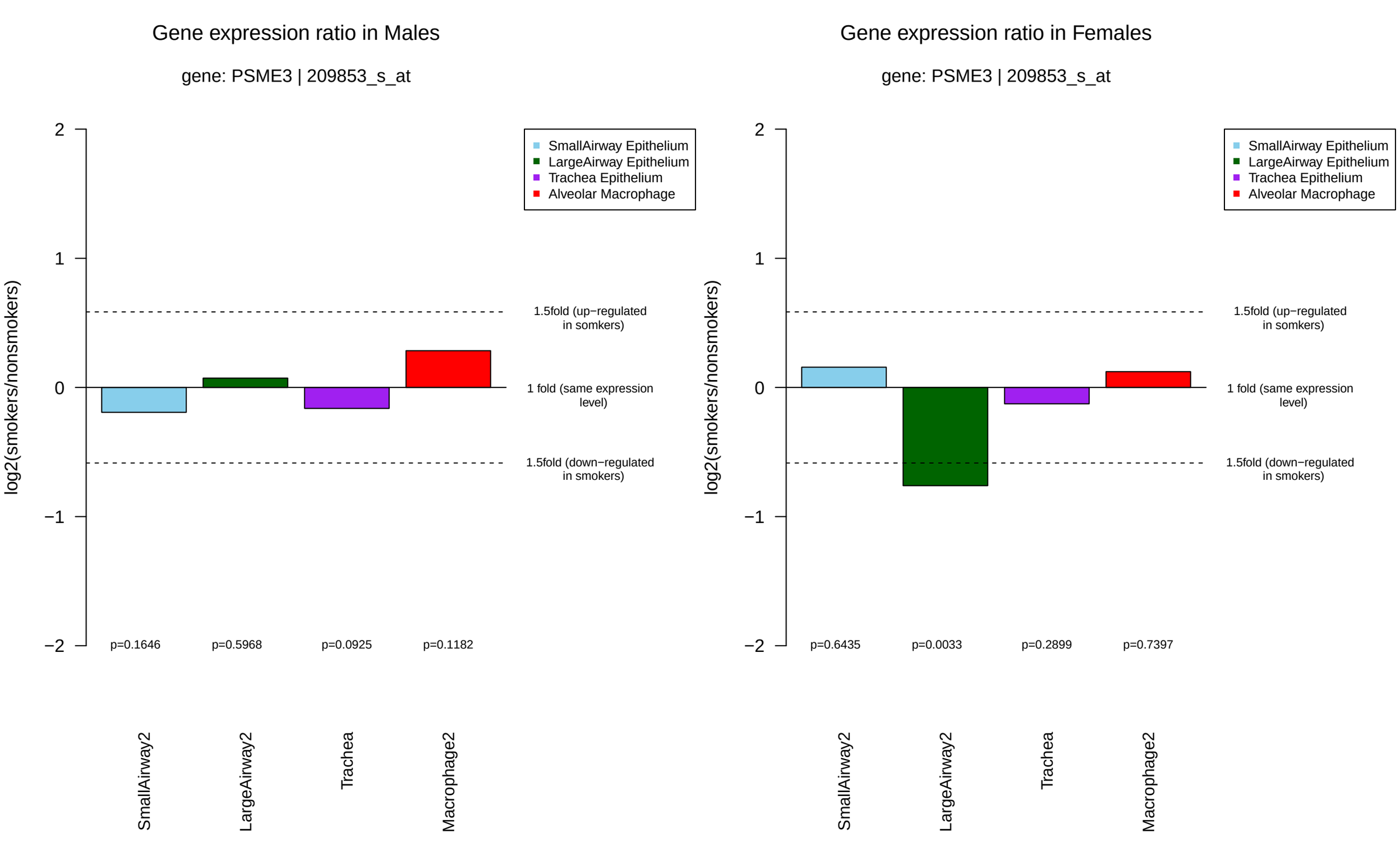
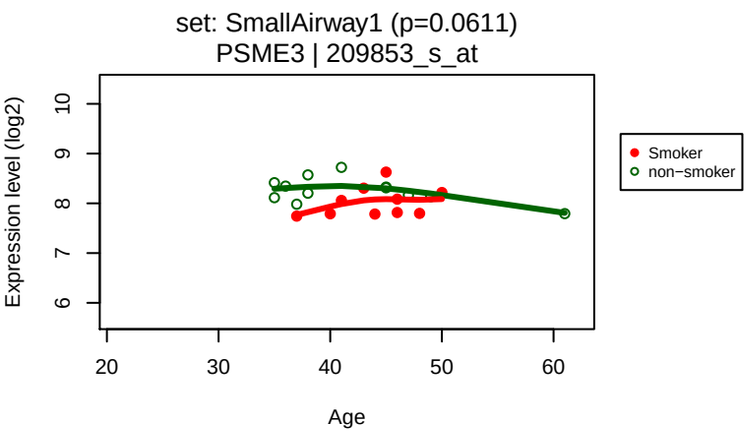
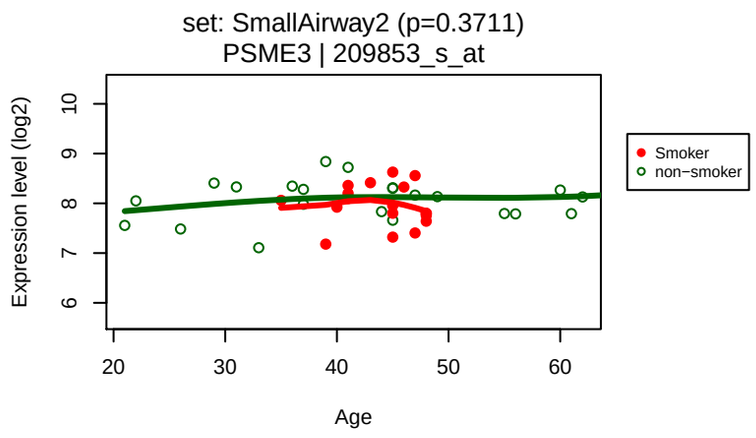
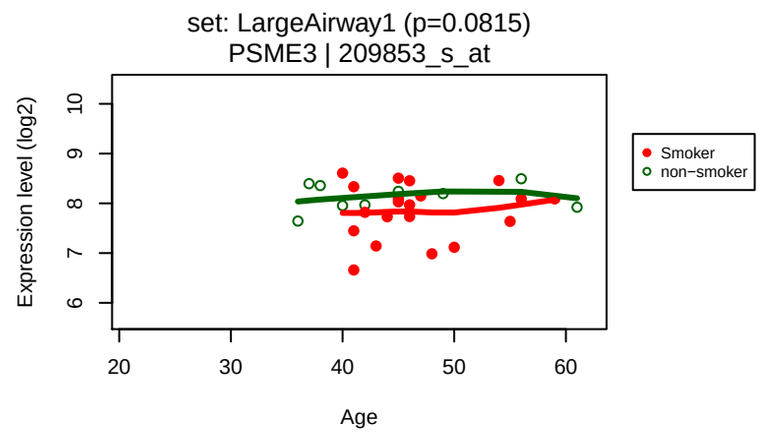

Gene expression of smokers and nonsmokers
gene: PSME3 | 209853_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0611
p=0.3711
p=0.4452
p=0.0815
p=0.0987
p=0.0446
p=0.0952
p=0.1552
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

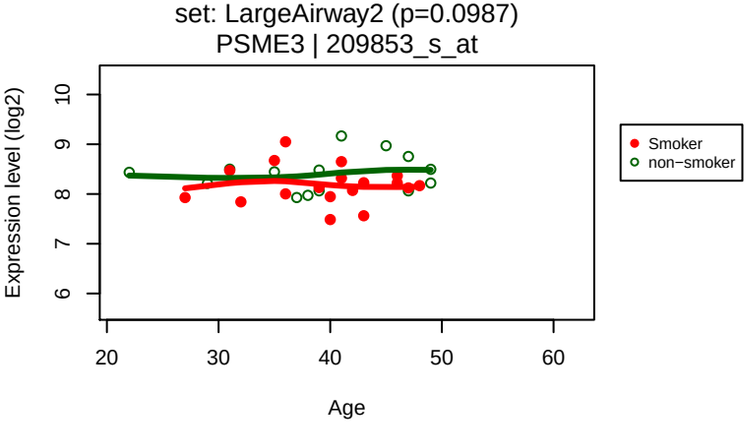

Gene expression ratio (smokers/nonsmokers)
gene: PSME3 | 209853_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0611
p=0.3711
p=0.4452
p=0.0815
p=0.0987
p=0.0446
p=0.0952
p=0.1552
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

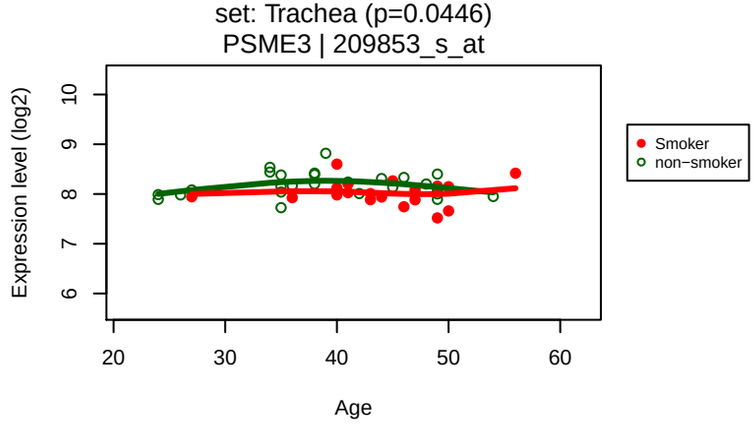

Gene expression ratio in Males
gene: PSME3 | 209853_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1646
p=0.5968
p=0.0925
p=0.1182
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PSME3 | 209853_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6435
p=0.0033
p=0.2899
p=0.7397
SmallAirway2
LargeAirway2
Trachea
Macrophage2

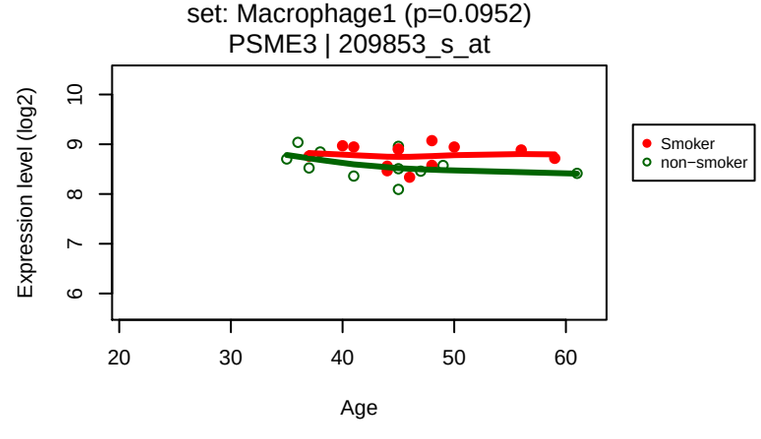

set: SmallAirway1 (p=0.0611)
PSME3 | 209853_s_at
Expression level (log2)
Age
Smoker
non−smoker

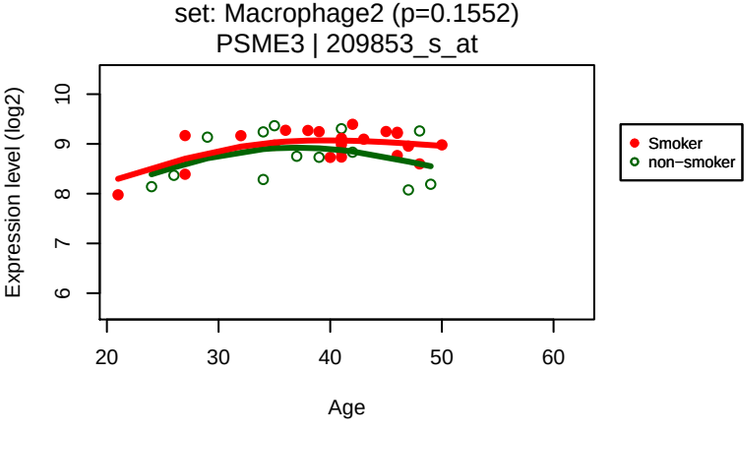

set: SmallAirway2 (p=0.3711)
PSME3 | 209853_s_at
Expression level (log2)
Age
Smoker
non−smoker

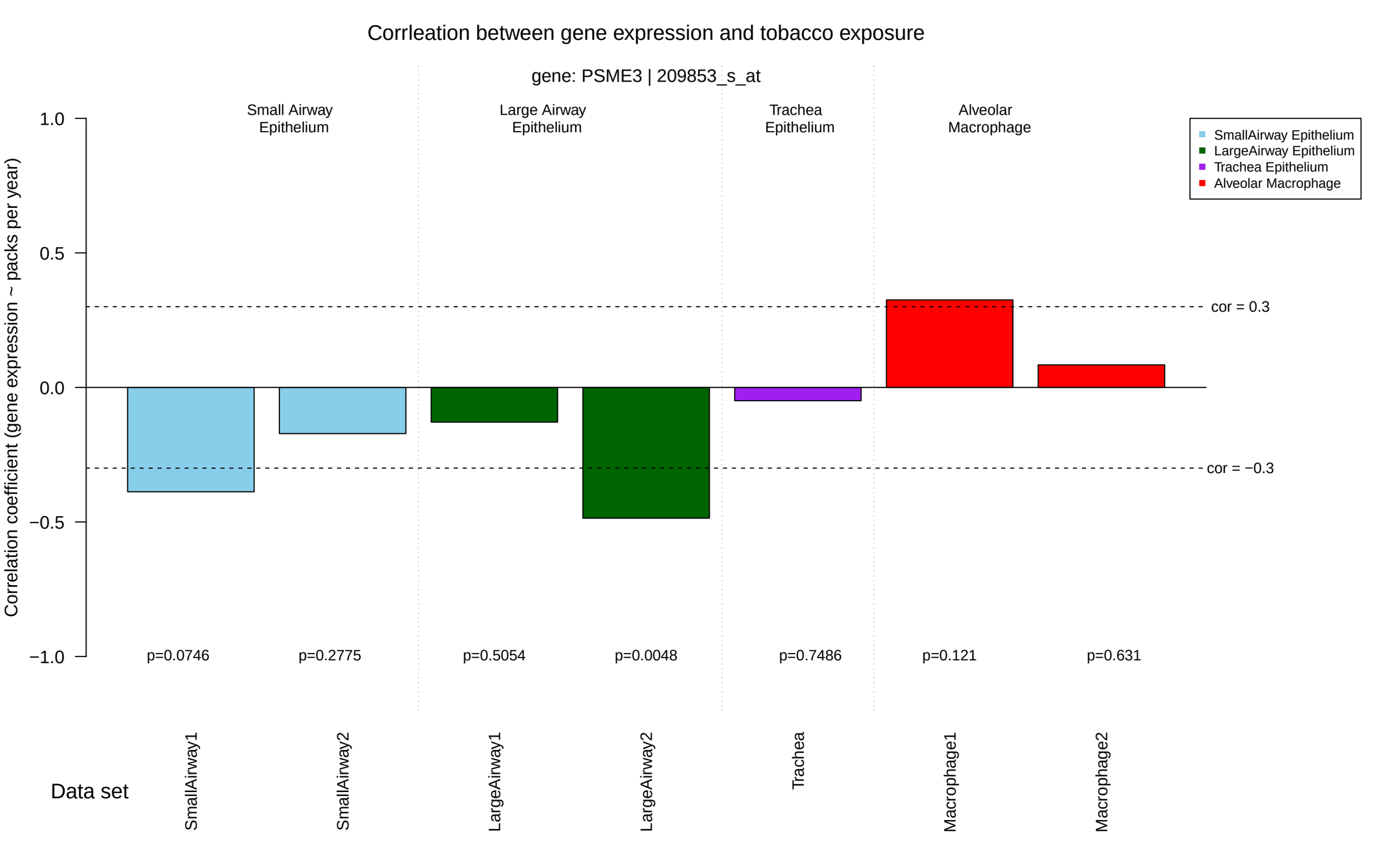

set: LargeAirway1 (p=0.0815)
PSME3 | 209853_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0987)
PSME3 | 209853_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0446)
PSME3 | 209853_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0952)
PSME3 | 209853_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1552)
PSME3 | 209853_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PSME3 | 209853_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0746
p=0.2775
p=0.5054
p=0.0048
p=0.7486
p=0.121
p=0.631
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2